3.0 Review methodology

To develop a comprehensive understanding of quantum machine learning and its application in the financial sector's cybersecurity domain, this study employed a systematic literature review. This method offers a structured framework for identifying, synthesizing, and critically evaluating existing research (Xiao & Watson, 2019). To enhance the credibility and rigor of the review, the process adhered to established methodological steps, including the formulation of a review protocol, systematic search of publications, screening of relevant literature, quality assessment of selected studies, data extraction, and synthesis of findings (Okoli, 2015). The subsequent subsections detail each of these steps.

3.1 Research Strategy

IEEE Explorer, Science Direct, Elsevier and Google Scholar, were the main databases used to explore and obtain publications used in this research. The selection of these search systems was because they publish vast academic works that are relevant to the current review, including literature reviews in the field of quantum machine learning, and cybersecurity in different sectors including financial applications. Google Scholar specifically was used to gather and aggregate content from different sources in QML, cybersecurity in the finance sector because of its accessibility, breadth, and speed. The search phrases used to search for the articles were aligned to the prevalent thematic areas of this field of study (Bunescu L, Vârtei AM (2024)). The quantum machine learning literature review by Pradeep et al. (2025) contributed to the keywords while the study Quantum Machine Learning for Anomaly Detection (Sounak et al., 2024) supplemented sustainability keywords. **Table 1** lists all the keywords applied in the search process.

The in-title operator was used in search databases to restrict results to articles with titles containing the exact search terms. In addition, OR and AND Boolean operators were used to ensure the presence of keywords in the text of the publications.

3.2 Search inclusion and exclusion criteria

This literature review focused on the intersection of quantum machine learning (QML), cybersecurity, and the financial sector. Given the emerging nature of quantum technologies and their increasing relevance to digital security, the review covered publications from **January 2015 to June 2025**, a period marked by significant advancements in quantum-enhanced algorithms and their applications in secure systems. This timeframe was selected to ensure that the study captured the most recent and impactful developments in the field.

To maintain a high standard of academic rigor, only peer-reviewed, open-access studies published in English were considered for inclusion. Gray literature, such as conference proceedings, theses, white papers, and institutional reports, was excluded due to concerns regarding consistency in methodological quality, peer-review rigor, and replicability. As noted by Xiao and Watson (2019), journal articles, books, and book chapters that undergo formal peer review are more likely to contribute substantively to systematic synthesis and evidence-based conclusions. Therefore, this study prioritized peer-reviewed academic publications to ensure both credibility and reliability in the resulting analysis.

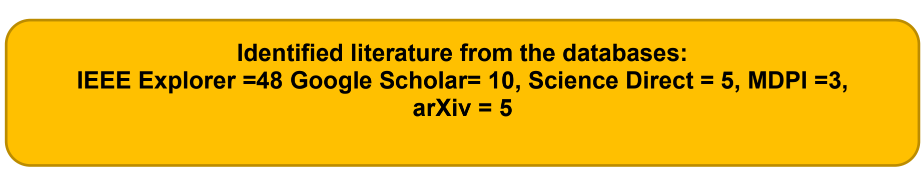
Table 1. Keywords applied in the search process

Concept Domain	**Keywords**
Quantum machine Leaning	Quantum Machine Learning OR "Quantum AI" OR "Quantum-enhanced Machine Learning" OR "Quantum Neural Networks" OR "Quantum Support Vector Machines" OR "Quantum Deep Learning" OR "Quantum Learning Algorithms"
Cybersecurity	(Cybersecurity OR "Cyber Security" OR "Information Security" OR "Fraud Detection" OR "Network Security" OR "Digital Security" OR "Cyber Defense" OR "Intrusion Detection" OR "Cyber Threats"
Financial Sector	Financial Sector OR "Banking Security" OR "Financial Services" OR "FinTech" OR "Digital Finance" OR "Financial Institutions" OR "Banking Cybersecurity" OR Finance "Fraud Detection" OR "Anomaly Detection" OR "KYC" OR "AML" OR "Authentication" OR "Financial Threat Detection" OR "Secure Transactions"

Source: Authors' own creation

3.3 Study selection criteria

Using the predefined keywords and Boolean search strings, publications were initially retrieved based on their alignment with the research questions (RQs) and adherence to the established inclusion and exclusion criteria. The initial search process yielded a total of 76 publications. Each entry was subjected to a **screening procedure** wherein the author examined the titles, abstracts, and keywords, followed by a rapid assessment of the full texts for those studies that appeared most relevant to the core themes of quantum machine learning (QML), cybersecurity, and financial sector applications. Through this process, 51 studies were excluded due to insufficient relevance or failure to meet the defined selection criteria. Consequently, 27 peer-reviewed journal articles were retained for in-depth review and synthesis. The detailed procedure of publication selection, including the screening and filtering steps, is illustrated in the flow diagram (**Figure 1**).



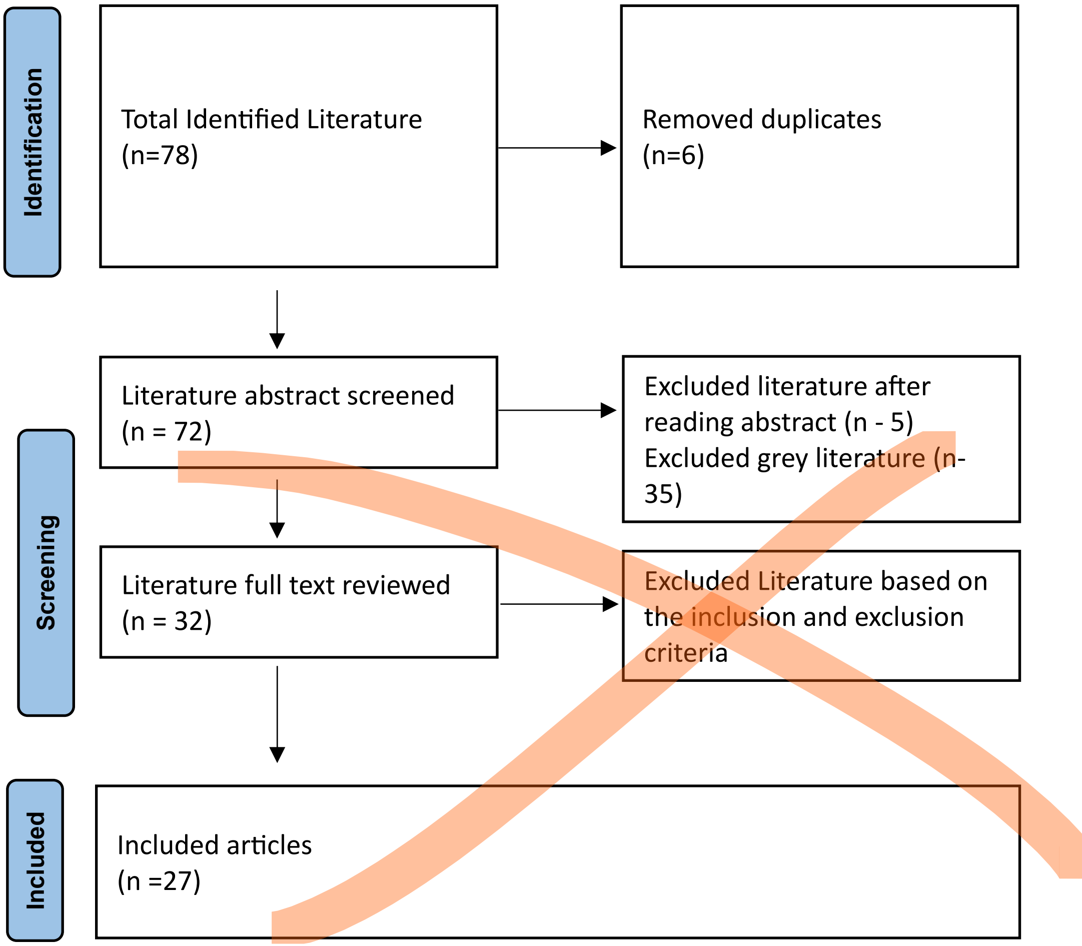


Figure 1. Identification, screening, evaluation for eligibility, and inclusion of literature according to PRISMA (Page et al, 2023)

3.4 Extraction of data

The primary objective of the data extraction phase was to systematically and accurately retrieve information from the selected publications that directly addressed the study's research questions (RQs). To ensure methodological rigor, the authors developed a structured coding framework that guided the extraction of relevant data in alignment with the RQs. In addition to answering the review questions, the extracted data encompassed key bibliographic and methodological descriptors, including title, database source, journal of publication, area of focus, research methodology, and year of publication.

To validate the reliability and internal consistency of the data extraction process, a test-retest approach was applied, whereby a second extraction was performed on a random subset of the primary studies. This procedure enabled the authors to assess the stability and reproducibility of the extracted data, thereby enhancing the credibility of the review findings (Kitchenham & Charters, 2007).

Table 2 presents the complete list of search string combinations that were employed during the database search process, providing transparency and replicability for future research efforts.

Table 2. Search string applied in the search process

Research Question	Search String
RQ1	(("Quantum Machine Learning" OR "Quantum AI" OR "Quantum-enhanced Machine Learning" OR "Quantum Neural Networks" OR "Quantum Support Vector Machines" OR "Quantum Deep Learning" OR "Quantum Learning Algorithms") AND (Cybersecurity OR "Cyber Security" OR "Information Security" OR "Fraud Detection" OR "Network Security" OR "Digital Security" OR "Cyber Defense" OR "Intrusion Detection" OR "Cyber Threats") AND ("Financial Sector" OR "Banking Security" OR "Financial Services" OR "FinTech" OR "Digital Finance" OR "Financial Institutions" OR "Banking Cybersecurity" OR Finance))
RQ2	("Quantum Machine Learning" OR "Quantum AI" OR "Quantum-enhanced Machine Learning" OR "Quantum Computing for AI") AND ("Financial Cybersecurity" OR "Banking Security" OR "Fraud Detection" OR "Anomaly Detection" OR "KYC" OR "AML" OR "Authentication" OR "Financial Threat Detection" OR "Secure Transactions")

Source: Authors' own creation

4. Results

This section addresses Research Questions 1 and 2 by presenting the analytical findings on the existing body of knowledge concerning the application of quantum machine learning (QML) in the context of cybersecurity within the financial sector. To respond comprehensively to these questions, the selected studies were systematically analyzed based on several key dimensions: authorship and institutional affiliations, temporal distribution of publications across the defined review period, stated research objectives, and contextual characteristics such as geographical focus and target population. In addition, attention was given to the research methodologies employed and the thematic orientations. This multi-dimensional analysis facilitates a nuanced understanding of how QML is being conceptualized and investigated in relation to cybersecurity challenges in finance.

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